Cover Page

Energize EV: Dynamic Wireless Power Transfer System

Theme: Environment and Sustainability

Team Name: ELECTRO

Team ID: HSC|TN|00501

State/UT: Tamil Nadu

Charge while moving. Reduce plug-in stops.

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Problem – Proposed Solution

THE PROBLEM

Charging downtime

EVs often need planned stops and plug-in access; charging infrastructure can be a barrier.

OUR PROPOSED SOLUTION

Transmit power from coils installed beneath a dedicated lane to a receiver coil under the EV.

HOW IT WORKS

1. **AC supply** →
2. **High-frequency inverter** →
3. **Lane transmitter coil** →
4. **Receiver coil** →
5. **Rectifier + battery**

Innovation: segmented coils energize only near the vehicle, with alignment and safety monitoring.

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Technical Approach

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POWER PATH

AC supply → inverter → transmitting coil → magnetic coupling → receiver coil → rectifier/filter → battery.

CONTROL & SAFETY

Controller monitors presence, alignment, temperature and power; shut down on abnormal conditions.

IMPLEMENTATION FLOW

Detect EV
Presence sensor

›

Align coils
Position check

›

Transfer
HF magnetic field

›

Condition
Rectify + filter

›

Monitor
Display status

Technology: inductive power transfer, MOSFET inverter, coils, rectifier, sensors, microcontroller and display.

Demo video: supplied separately with this submission. GitHub/GitLab: Not applicable (hardware-concept demo; no code repository submitted). AI tools: used for drafting/visual support; technical claims require team validation.

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Impact Potential

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ENVIRONMENTAL

Supports EV adoption and cleaner mobility when powered by low-carbon electricity.

USER BENEFIT

May reduce charging-stop time on equipped routes and improve convenience.

ECONOMIC POTENTIAL

Opportunity for smart-road infrastructure, service jobs and fleet charging models.

STAKEHOLDERS

EV users, fleet operators, road authorities, utilities, charging providers and city planners.

SDG alignment: SDG 7 (Affordable and Clean Energy) and SDG 11 (Sustainable Cities and Communities).

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Feasibility and Risk Analysis

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FEASIBILITY

Build a low-power tabletop demonstrator first; scale only after efficiency and safety testing.

CHALLENGES

Coil misalignment, transfer losses, heating, foreign objects, weather and installation cost.

MITIGATION

Use segmented activation, shielding, temperature sensing, current limits and emergency cutoff.

VALIDATION PLAN

Measure input/output power, coil temperature, alignment tolerance and charging stability.

Important: road deployment requires certified electrical safety, electromagnetic exposure checks, civil works, grid connection and standards compliance. This concept is not a claim of a certified road-ready charger.

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Team

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TEAM LEAD

DEEPIKHA SR

Coordination, planning and presentation.

TEAM MEMBER

ARASI L T

Research, technical explanation and documentation.

TEAM JOURNEY

- Defined the charging-access problem and mapped a coil-based solution.
- Combined EEE concepts: electromagnetic induction, power conversion and control.
- Next steps: assemble a safe low-voltage prototype, record measurements and improve coil alignment.

References

REFERENCE & RESEARCH BASIS

1. SAE International — J2954, Wireless Power Transfer for Light-Duty Plug-in/Electric Vehicles (standard overview).

2. U.S. Department of Energy, Alternative Fuels Data Center — Electric Vehicle Charging Basics.

3. International Energy Agency (IEA) — Global EV Outlook (EV adoption and charging infrastructure context).

4. Project team concept: Energize EV — dynamic inductive power transfer lane (proposed prototype concept).

Note: The concept requires experimental validation; no charging speed, efficiency or cost figure is claimed without measurements.